Module 12 Assignment — Arnav Kucheriya

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Q1) State any three disadvantages associated with the fixed-sized partitioning technique of memory management.

1. **Internal fragmentation**
   In fixed-sized partitioning, a process may not fully use the partition assigned to it. The unused portion of that partition is wasted.
2. **Limited number of processes in memory**
   Since memory is divided into a fixed number of partitions, only that many processes can be loaded at one time, even if some free space still exists.
3. **Process size limitation**
   A process larger than the size of any partition cannot be loaded into memory, even if the total free memory across partitions would be enough.

Q2) Dynamic Partitioning
Given Memory Configuration
The free memory blocks, from left to right, are:

- 5M
- 3M
- 4M
- 6M
- 2M
- 4M
- 3M

The last process placed is marked with **X**, and a new **2-MB** allocation request must be satisfied.

A) Best Fit
Best Fit chooses the **smallest free block** that can satisfy the request.
Available free blocks:

- 5M
- 3M
- 4M
- 6M
- 2M
- 4M
- 3M

The smallest block that can hold 2M is the **2M block** itself.
Result

- The new process is allocated in the 2M block.
- The entire block is used.
- No free space remains in that location.

Memory Strip
[Used][5M free][Used][3M free][Used][4M free][Used][6M free][X][2M allocated][Used][4M free][Used][3M free]

B) First Fit
First Fit chooses the **first free block from the left** that is large enough.
Scanning from left to right:

- 5M -> fits 2M

Result

- The new process is allocated in the **first 5M block**.
- 2M is used.
- 3M remains free.

Memory Strip
[Used][2M allocated][3M free][Used][3M free][Used][4M free][Used][6M free][X][2M free][Used][4M free][Used][3M free]

C) Next Fit
Next Fit begins searching **from the location of the last placement**, which is marked by **X**.
Starting from X and moving right:

- The next free block is 2M
- It fits exactly

Result

- The new process is allocated in the 2M block immediately after X.
- The entire block is used.

Memory Strip
[Used][5M free][Used][3M free][Used][4M free][Used][6M free][X][2M allocated][Used][4M free][Used][3M free]

Final Answer for Q2

- **Best Fit:** allocate in the **2M block**
- **First Fit:** allocate in the **first 5M block**
- **Next Fit:** allocate in the **2M block immediately after X**

Q3) Buddy System
A 2-Mbyte block of memory is allocated using the buddy system.
Total Memory
2 MB = 2048 KB
In the buddy system, each request is allocated the **smallest power-of-two block** that can hold it.

Q3(A) Show the results of the following sequence by dividing memory into appropriate blocks.
Initial State
[2048 free]

1. A requests 156 KB
The smallest power of 2 greater than or equal to 156 is:
256 KB
Split as needed:
2048 -> 1024 + 1024
1024 -> 512 + 512
512 -> 256 + 256
Allocate one 256 KB block to A.
State
[A256][256 free][512 free][1024 free]

2. B requests 448 KB

The smallest power of 2 greater than or equal to 448 is:
512 KB
Allocate one 512 KB block to B.
State
[A256][256 free][B512][1024 free]

3. C requests 511 KB
The smallest power of 2 greater than or equal to 511 is:
512 KB
Split the 1024 KB free block:
1024 -> 512 + 512
Allocate one 512 KB block to C.
State
[A256][256 free][B512][C512][512 free]

4. Release B
B’s 512 KB block becomes free.
State
[A256][256 free][512 free][C512][512 free]

5. D requests 259 KB
The smallest power of 2 greater than or equal to 259 is:
512 KB
Allocate the free 512 KB block to D.
State
[A256][256 free][D512][C512][512 free]

6. E requests 97 KB
The smallest power of 2 greater than or equal to 97 is:
128 KB
Split the remaining 512 KB block:
512 -> 256 + 256
256 -> 128 + 128
Allocate one 128 KB block to E.
State
[A256][256 free][D512][C512][E128][128 free][256 free]

7. F requests 75 KB
The smallest power of 2 greater than or equal to 75 is:
128 KB
Allocate the remaining 128 KB block to F.
State
[A256][256 free][D512][C512][E128][F128][256 free]

8. Release C
C’s 512 KB block becomes free.
State
[A256][256 free][D512][512 free][E128][F128][256 free]

9. Release F
F’s 128 KB block becomes free.
State
[A256][256 free][D512][512 free][E128][128 free][256 free]

10. Release A
A’s 256 KB block becomes free.
Its buddy 256 KB block is also free, so they merge:
256 + 256 -> 512 free
State
[512 free][D512][512 free][E128][128 free][256 free]

11. Release D
D’s 512 KB block becomes free.
Its buddy 512 KB block is also free, so they merge:
512 + 512 -> 1024 free
State
[1024 free][512 free][E128][128 free][256 free]

12. Release E
E’s 128 KB block becomes free.
Now merge repeatedly:
128 + 128 -> 256 free
256 + 256 -> 512 free
512 + 512 -> 1024 free
1024 + 1024 -> 2048 free
Final State
[2048 free]

Q3(A) Summary of All States
Initial:
[2048 free]
After A:
[A256][256 free][512 free][1024 free]
After B:
[A256][256 free][B512][1024 free]
After C:
[A256][256 free][B512][C512][512 free]
After Release B:
[A256][256 free][512 free][C512][512 free]
After D:
[A256][256 free][D512][C512][512 free]
After E:
[A256][256 free][D512][C512][E128][128 free][256 free]
After F:
[A256][256 free][D512][C512][E128][F128][256 free]
After Release C:
[A256][256 free][D512][512 free][E128][F128][256 free]
After Release F:
[A256][256 free][D512][512 free][E128][128 free][256 free]
After Release A:
[512 free][D512][512 free][E128][128 free][256 free]
After Release D:
[1024 free][512 free][E128][128 free][256 free]
After Release E:
[2048 free]

Q3(B) Show the binary tree representation following Release C
After Release C, the memory strip is:

[A256][256 free][D512][512 free][E128][F128][256 free]

Binary Tree Representation

```
2048
├── 1024
│   ├── 512
│   │   ├── 256 (A)
│   │   └── 256 (Free)
│   └── 512 (D)
└── 1024
    ├── 512 (Free)
    └── 512
        ├── 256
        │   ├── 128 (E)
        │   └── 128 (F)
        └── 256 (Free)
```

Correspondence of Leaf Nodes to Memory Strip

0–255 -> A
256–511 -> Free
512–1023 -> D
1024–1535 -> Free
1536–1663 -> E
1664–1791 -> F
1792–2047 -> Free

Q4) Paging

Given

- Program size = 256 bytes
- Page size = 16 bytes
- Memory is 2-byte addressable
- Main memory is divided into 32 frames

Page Table

Page No.	Frame No.
0	14
1	22
2	10
3	17
4	12
5	9
6	31
7	---
8	---
9	---
10	---
11	---

12	---
13	---
14	---
15	---

Important Step
Since memory is **2-byte addressable**, one address refers to 2 bytes.
So the number of addressable locations in one page is:
Page size / addressable unit = 16 / 2 = 8
Thus:

- **1 page = 8 addressable locations**

Q4(A) Convert Logical Address 49 to Physical Address
Step 1: Find the page number
Page number = floor(49 / 8) = 6

Step 2: Find the offset
Offset = 49 mod 8 = 1
So:
Logical address 49 = Page 6, Offset 1

Step 3: Find the frame number from the page table
From the page table:
Page 6 -> Frame 31

Step 4: Compute the physical address
Physical address = (Frame number × page size in addressable locations) + offset
= (31 × 8) + 1
= 248 + 1
= 249
Logical address 49 -> Physical address 249

Q4(B) Convert Logical Address 22 to Physical Address
Step 1: Find the page number
Page number = floor(22 / 8) = 2

Step 2: Find the offset
Offset = 22 mod 8 = 6
So:
Logical address 22 = Page 2, Offset 6

Step 3: Find the frame number from the page table
From the page table:
Page 2 -> Frame 10

Step 4: Compute the physical address
Physical address = (Frame number × page size in addressable locations) + offset
= (10 × 8) + 6
= 80 + 6
= 86
Logical address 22 -> Physical address 86